Learning and compressing URA codebooks

Evidence, limitations and the next research question

1. What does the explicit reference buy us?

Aim. Find a compact encoder and a tractable multiuser decoder that retain the performance of a broad, explicit codebook. Small payloads let us separate encoder restrictions from receiver limitations; they do not establish competitiveness at the original papers' dimensions.

Common experiment. Real, synchronous, unfaded AWGN; known K; $M = 2^B$ messages:

$$y = \Phi a + z, \quad \Phi \in \mathbb{R}^{n \times M}, \quad \|\phi_j\|_2 = 1, \quad z \sim \mathcal{N}(0, \sigma^2 I), \quad \sigma^2 = \frac{1}{2B(E_b/N_0)}.$$

The primary test draws K distinct messages: $a \in \{0,1\}^M$, $\mathbf{1}^T a = K$. PUPE is $\mathbb{E}[|\mathcal{S} \setminus \widehat{\mathcal{S}}|/K]$, with at most K decoded identities. We use $B \in \{12, 14\}$, $n = 256$, $K \in \{7, 15, 22, 26\}$ and $E_b/N_0 = -4{:}2{:}10$ dB. Each point averages three seeds, each with 256 frames. "Fixed" means the encoder is frozen; D2 is trained. Native comparators use ODMA placement plus polar-list decoding, or coded compressed sensing with AMP/BP. Replacing their receiver with D2 tests the codebook separately from its native inference procedure.

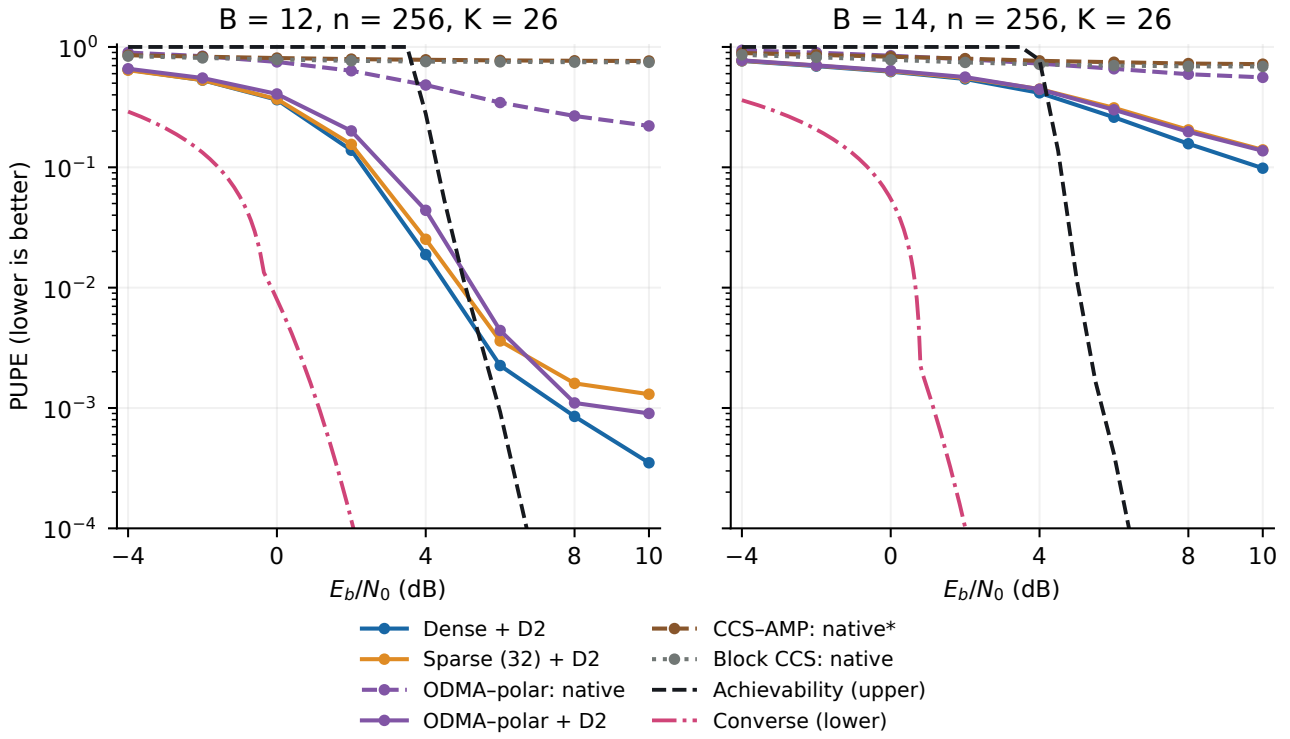


Figure 1: High-load comparison: fixed codebooks, distinct messages, three-seed means. Published schemes are adapted to these small payloads. *CCS–AMP uses nominal, not per-codeword peak, energy; its curve is contextual rather than a strictly energy-matched comparison.

- **The receiver explains most of the original ODMA gap.** At $B = 14, K = 26, 10$ dB, native ODMA–polar has PUPE 0.559; the *same codebook* with D2 gives 0.137, versus 0.098 for dense D2. Receiver replacement closes 92% of that gap (95.5% over the full tested grid).
- **A smaller encoder gap remains.** Full-grid means at $B = 14$: dense 0.285, random sparse 0.299, ODMA–polar 0.296 under D2. Similar results do not imply equivalent or optimal codebooks.
- **Read the bounds as a bracket, not a target curve.** $\epsilon_{\text{converse}} \le \epsilon^* \le \epsilon_{\text{ach}}$, where ϵ^* is the best achievable PUPE. A method may beat the achievability curve, but not a valid converse in expectation. Bound definitions are on page 4.

2. Can the encoder be generated instead of stored?

Starting point. Dense storage is $n2^B$ reals; sparse storage is $T2^B$ amplitudes plus supports. Earlier sparsity sweeps motivated compression. Here $T = 32$ is 12.5% density, more aggressive than the original 25% test.

Current construction: one nonzero per table, coordinate-wise amplitude banks. Partition $n = TR$ coordinates into T tables of $R = 2^r$ positions. For message $w \in \mathbb{F}_2^B$,

$$
\begin{aligned}
h_t(w) &= \mathrm{int}(A_t w \oplus b_t), & A_t &\in \mathbb{F}_2^{r\times B}, \quad b_t \in \mathbb{F}_2^r,\\
u_t(w) &= \mathrm{int}(P_t w), & P_t &\in \mathbb{F}_2^{J\times B}, \quad V \in \mathbb{R}^{T\times 2^J},\\
g_t(w) &= V_{t,u_t(w)}, & \alpha(w) &= g(w)/\|g(w)\|_2,\\
\phi(w) &= \sum_{t=0}^{T-1} \alpha_t(w) e_{tR+h_t(w)}.
\end{aligned}
$$

Indices are zero-based. Binary maps choose *where* and *which bank entry*; amplitudes are real, not bits. Disjoint tables give $\|\phi(w)\|_2 = 1$ for every nonzero $g(w)$. Full-rank stacked hashes ensure distinct support tuples. Maps stay fixed; joint training updates V through lookup, normalization and the decoder.

Each table has its *own* label u_t; this is not reuse of one complete amplitude vector across messages. At $J = B$, invertible P_t recovers arbitrary amplitudes on the chosen supports. At $B = 14, T = 32$:

$$
\underbrace{524{,}288}_{\text{unrestricted}} \longrightarrow \underbrace{32{,}768}_{J=10\ (16\times)} \longrightarrow \underbrace{8{,}192}_{J=8\ (64\times)} \longrightarrow \underbrace{512}_{J=4\ (1{,}024\times)} \quad \text{stored amplitudes.}
$$

Map metadata adds $O(T(r + J)B)$ bits. With bounded J, generation is no longer exponential in B; decoding still is.

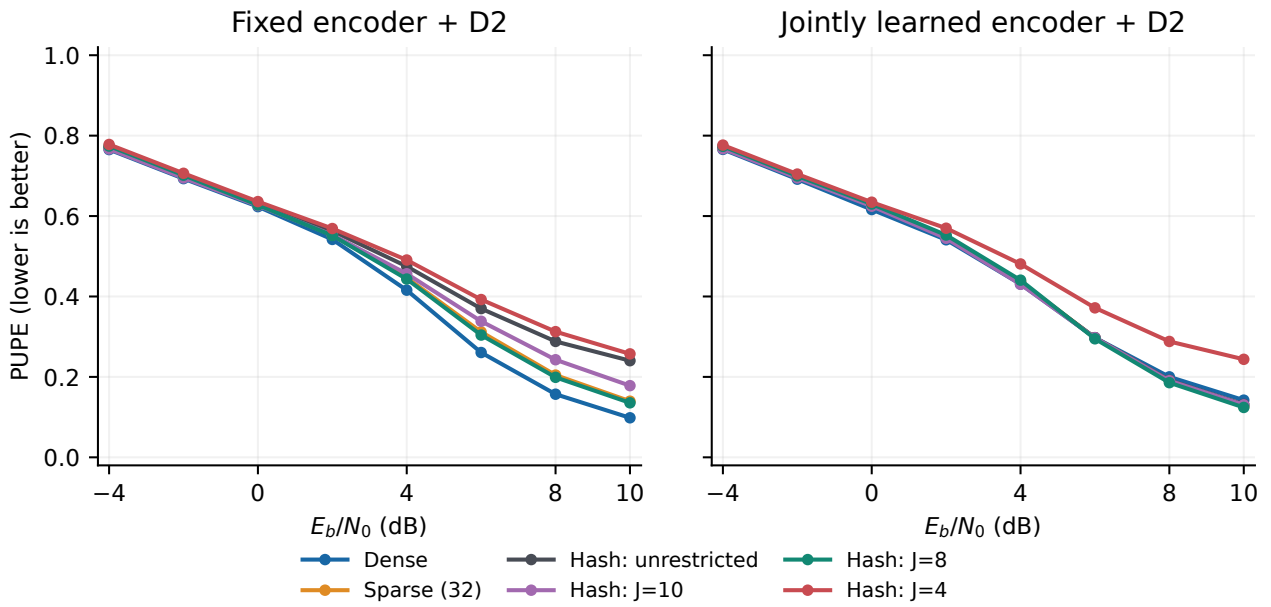


Figure 2: $B = 14, n = 256, K = 26$: same support hashes across amplitude families. Both panels train D2; the right also learns the encoder. Some runs were training-sensitive, so the ordering is not a clean ranking of encoder model classes.

- $J = 8$ **is a promising compression point.** Joint $J = 8$ and unrestricted hash have almost identical full-grid means, 0.29361 and 0.29331, with 64× fewer amplitudes. Three seeds do not certify equivalence. Some $J = 4$ and unrestricted runs were unstable; initialization and training need improvement before declaring the strongest compression harmful. Performance at larger B remains untested.
- **Hashing is not uniquely favoured.** Current ODMA–polar is similarly competitive under D2 and already has a structured inverse. Its 64/256 placement patterns at $B = 12/14$ differ from the poor historical four-mask control.
- **The missing step is inversion.** These tests score all 2^B messages. Our earlier sectioned parity/BP trials failed to recover users' messages at large B: fragment association remained unresolved. This does not rule out sectioned codes generally.

3. What changed in the decoder?

	State	Main idea and present limitation
D0	2^B	Unrolled residual updates; learned scalar variance/evidence calibration.
D1	2^B	D0 plus learned global context; more parameters do not ensure better inference.
D2	2^B	Geometry-based interference estimate, likelihood update and known-K projection.
D3	$Q = 1024$	D2 on a proposed list; faster refinement, but missed candidates are unrecoverable.
D4	$Q = 1024$	D3 plus learned pairwise candidate interactions; results still incomplete.

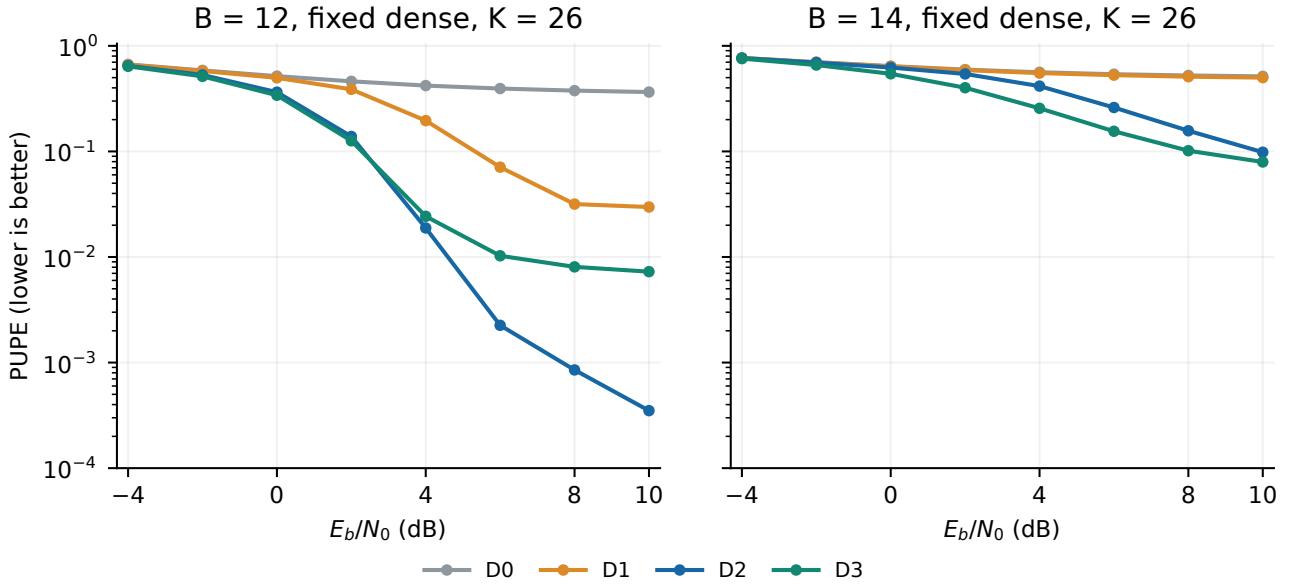


Figure 3: Same fixed dense codebooks, eight layers, same evaluation conditions. D2 makes a large gain over D0/D1; restricting to candidates can help or hurt. D4 is omitted rather than averaging incomplete groups.

D2's rationale. Let $G = \Phi^T\Phi$, $d_j = G_{jj}$ and $s_j = \sum_k G_{jk}^2$. For residual $r^t = y - \Phi a^t$, adding back the self-term gives an exact decomposition:

$$\zeta_j^t = \phi_j^T r^t + d_j a_j^t = d_j a_j^\star + \underbrace{\sum_{k\neq j} G_{jk}(a_k^\star - a_k^t) + \phi_j^T z}_{\text{interference plus noise}}.$$

Approximating this interference as Gaussian with mean-field variance gives

$$\widehat{\omega}_{j,t} = c_t\big[\sigma^2 d_j + \bar{v}_{-j}^t(s_j - d_j^2)\big], \qquad v_k^t = a_k^t(1-a_k^t), \quad \bar{v}_{-j}^t = \frac{\sum_{k\neq j} v_k^t}{M-1}.$$

The resulting active-versus-inactive evidence and cardinality projection are

$$e_{j,t} = \frac{d_j(\zeta_j^t - d_j/2)}{\widehat{\omega}_{j,t}}, \qquad p_j^t = \mathrm{sigmoid}(e_{j,t} + \lambda_t), \quad \sum_j p_j^t = K, \qquad a^{t+1} = \beta_t a^t + (1-\beta_t)p^t.$$

Only the variance scale $c_t > 0$ and damping $\beta_t \in (0,1)$ learn: **16 scalars for eight layers**. Codebook geometry enters analytically; the uniform prior is absorbed into λ_t.

- **Useful structure, not a posterior guarantee.** D2 accounts for column correlations and separates evidence from cardinality. It is still an approximate, repeatedly reused effective channel: no Onsager correction or state-evolution claim. Changed losses/calibration accompany the change, so no single ingredient is causally isolated.
- **D3 exposes the proposal trade-off.** At $B = 14, K = 26$, 10 dB, dense D3 gives 0.0794 versus D2's 0.0983, despite a proposal miss rate near 0.0533. At $B = 12$, D3's 0.00726 error is already its proposal floor, versus D2's 0.00035. Current proposals rank all messages by matched-filter evidence: still exponential work.
- **More learning is not automatically better.** D1's extra capacity and joint encoder learning do not consistently improve the results. D2 is a strong working reference, not an optimal decoder; D4's ranking is still pending.

4. Research direction: locate the gain, then make it tractable

Where can we gain performance, and how much survives a tractable encoder and receiver? The explicit small-B system helps separate these questions; it is not the intended final architecture.

1. **Encoding or inference: where is the remaining opportunity?** Compare established and generated codebooks under the same strong receivers. If their performance stays close as inference improves, prioritize better decoding of a structured code. If an encoder advantage persists across receivers, identify the geometry responsible and preserve it in a compact construction. The present ODMA result favours investigating the first possibility.
2. **What actually benefits from learning?** Separate analytical inference, learned decoder calibration, and joint encoder–decoder learning. If most gains come from a small learned correction, retain the analytical structure. If learned encoders give a reproducible additional gain, ask which part of that adaptation can be generated without a parameter for every message. Greater expressivity alone is not the objective.
3. **Which useful structure also admits an inverse?** Treat ODMA–polar and generated hashes as alternative starting points. Polar constraints offer an existing search mechanism; hashes offer flexible compact codebooks. Compare full-alphabet recovery with bounded-search recovery on the *same* encoder. If the loss appears only after restricting search, improve the inverse rather than assuming the encoder is inadequate.
4. **Can the gain survive scaling?** Carry promising encoder/receiver combinations to larger payloads and paper-scale comparisons. Require candidate recall, memory and runtime alongside PUPE: the goal is to close the performance gap without enumerating 2^B messages. A small-B gain motivates this step; it does not predict its outcome.

Working hypothesis. Much of the useful freedom may already exist in tractable code constructions; the larger opportunity may be recovering their messages better. Hash/amplitude generation remains a credible alternative, not a conclusion we need to defend. The research should determine which combination of structure, inference and learning offers the best performance–complexity trade-off.

What exactly are the two bounds?

The black curve is Polyanskiy's Gaussian-MAC achievability reference [1], an *upper bound on optimal PUPE*. It uses the original numerical recipe (all p_t terms and q_1), omitting the collision penalty for distinct messages. The survey's Gaussian reference is this same result, not a second independent bound.

The lower curve takes the stronger of a genie-aided binary-testing/list converse and a finite-set rate-distortion converse combined with channel capacity [2]. It uses the same distinct-message, known-K, real-AWGN, list-size-K model. Both bounds allow column energy ≤ 1; our matched codebooks have energy 1. Finite-M set sizes are retained rather than assuming B is large. For $B = 14, n = 256, K = 26$, target PUPE 0.05 is bracketed by roughly **0.07 dB necessary** and **4.68 dB sufficient**; none of the principal production D2 means reaches it by 10 dB. This leaves room for improvement, but neither bound diagnoses encoder versus decoder loss.

Fair-comparison and evidence limits

- **Small-B adaptations, not native-scale paper reproductions.** ODMA–polar [3] uses shortened polar codewords; CCS–AMP/BP [4] uses a small parity graph. Block CCS is a variant, not a third independent paper.
- **CCS has an additional obstruction here.** The small-B triad admits distinct message sets with exactly the same noiseless sum; changing the decoder cannot resolve these ambiguities. Its poor results do not establish a limitation of the original large-B construction. Nominal-energy CCS is not peak-energy matched.
- **Measured, not proved optimal.** Complete three-seed groups are compared; the latest matched batch is fully returned. Training sensitivity limits fine encoder rankings. No result yet certifies equivalence, optimality or scalable recovery.

References and reproducibility. Frozen data, source hashes, the figure builder and audit links accompany this report.

[1] Y. Polyanskiy, A perspective on massive random-access, ISIT 2017.

[2] J.-F. Chamberland, K. Narayanan and Y. Polyanskiy, Unsourced Multiple Access: Information Theory and Coding, ISIT tutorial, 2021.

[3] M. Özates, M. Kazemi and T. M. Duman, Unsourced Random Access Using ODMA and Polar Codes, IEEE WCL, 2024.

[4] V. K. Amalladinne et al., Unsourced Random Access with Coded Compressed Sensing: Integrating AMP and Belief Propagation, IEEE TIT, 2022.